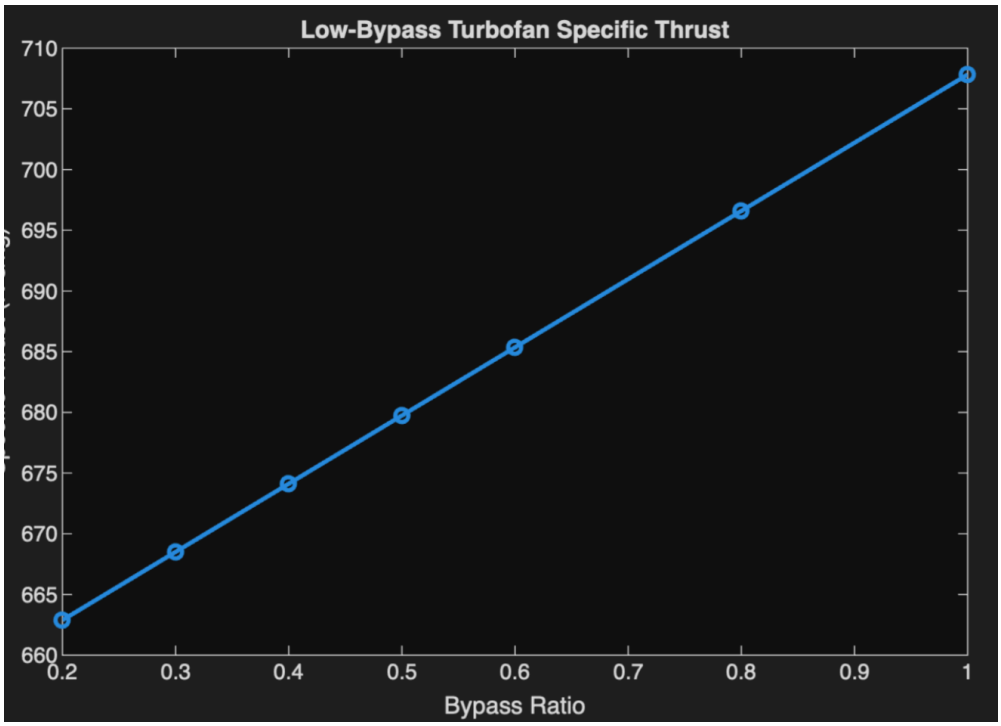
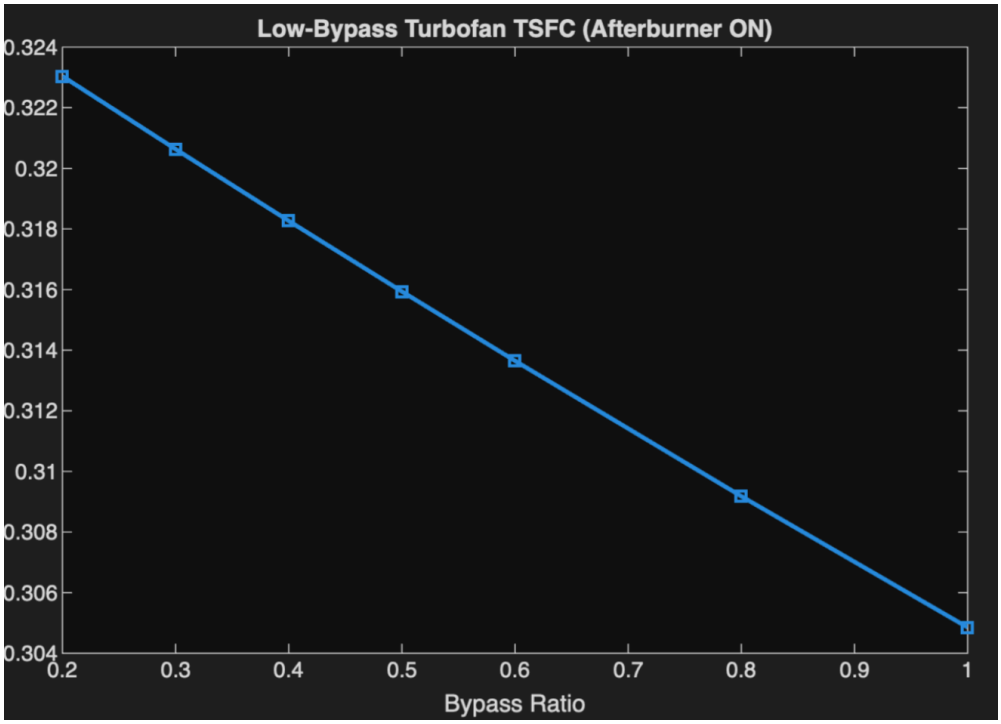
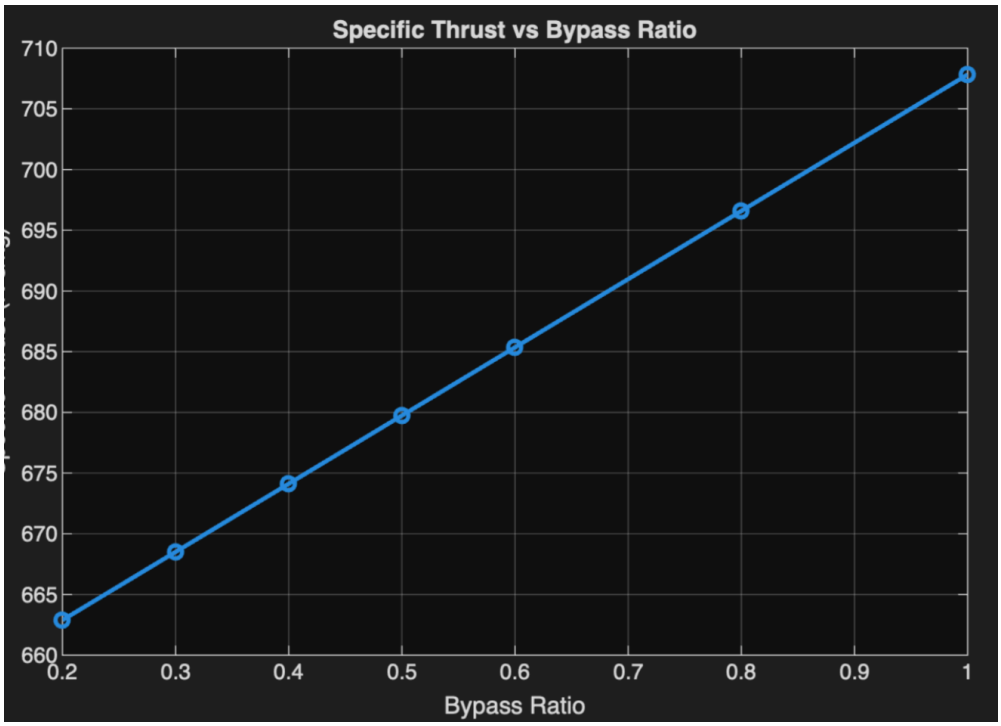
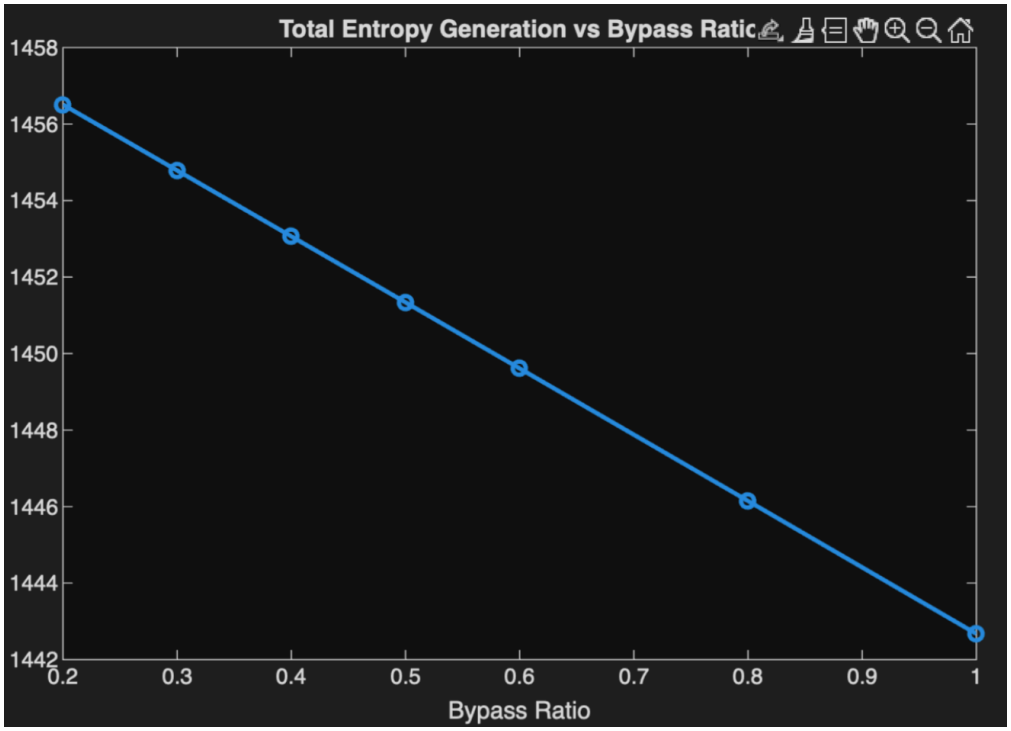
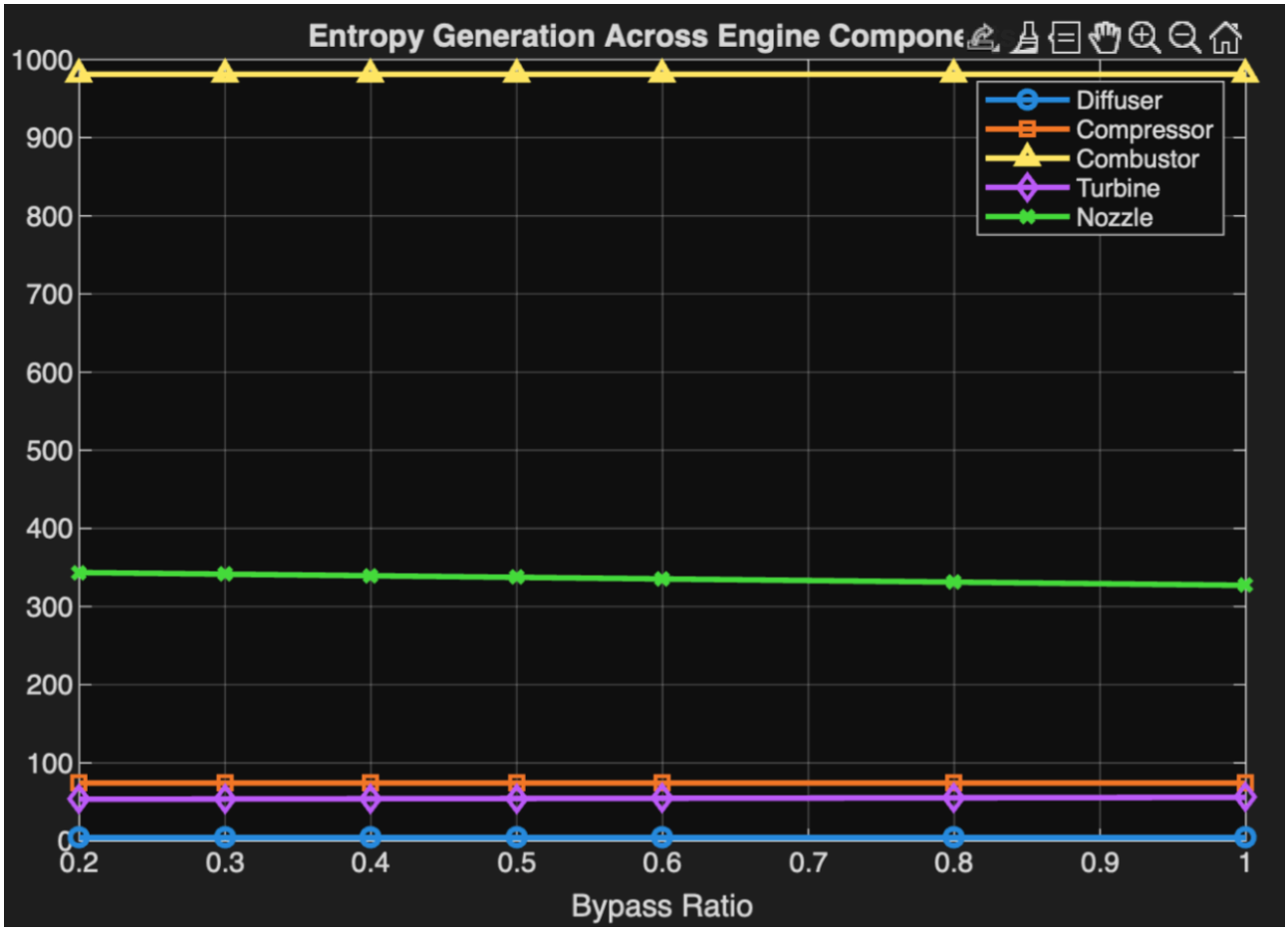
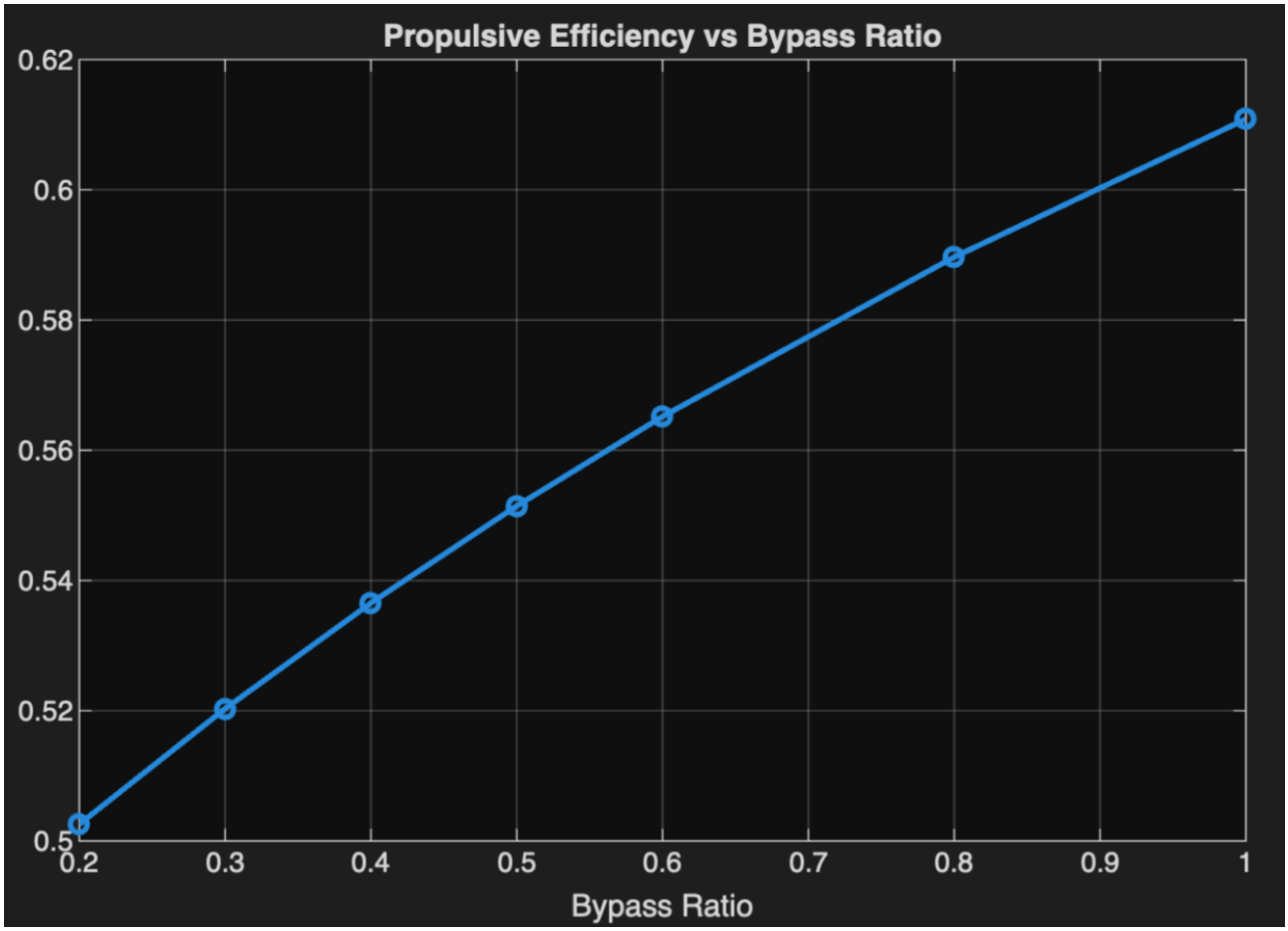

Low-Bypass Turbofan Specific Thrust
710
705
700
695
690
685
680
675
670
665
660
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio

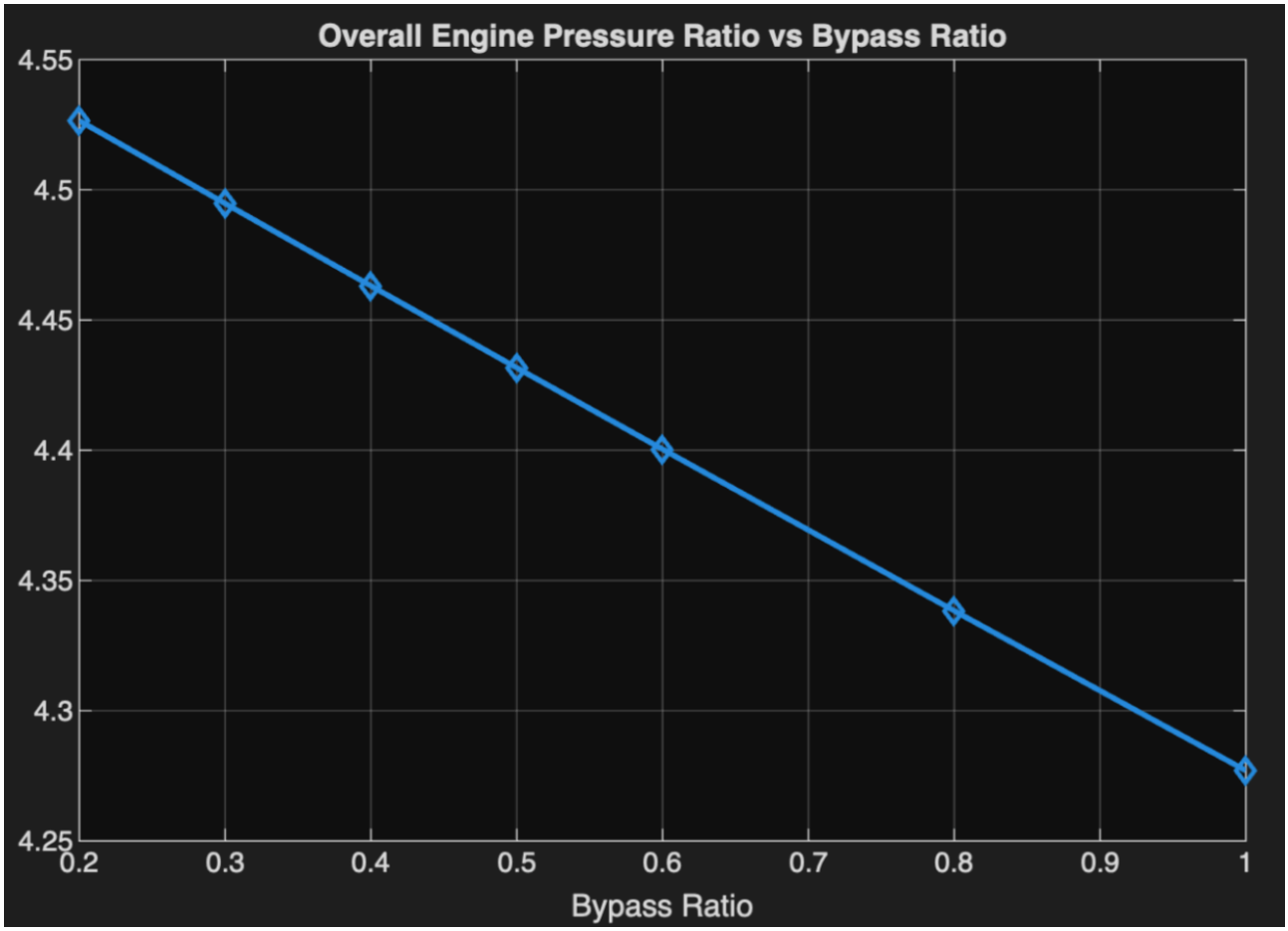

Low-Bypass Turbofan TSFC (Afterburner ON)
0.324
0.322
0.32
0.318
0.316
0.314
0.312
0.31
0.308
0.306
0.304
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio

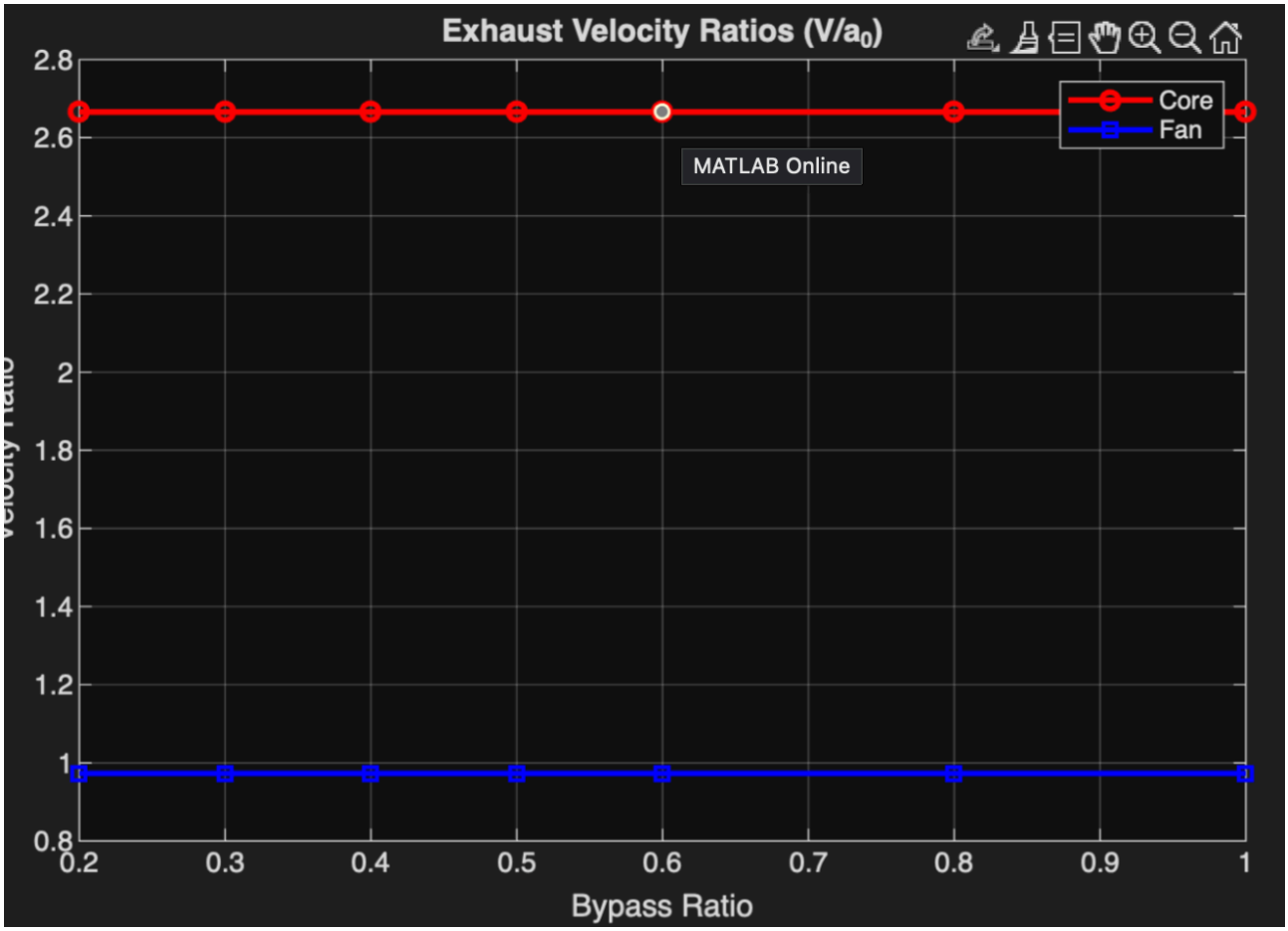

Specific Thrust vs Bypass Ratio
710
705
700
695
690
685
680
675
670
665
660
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio

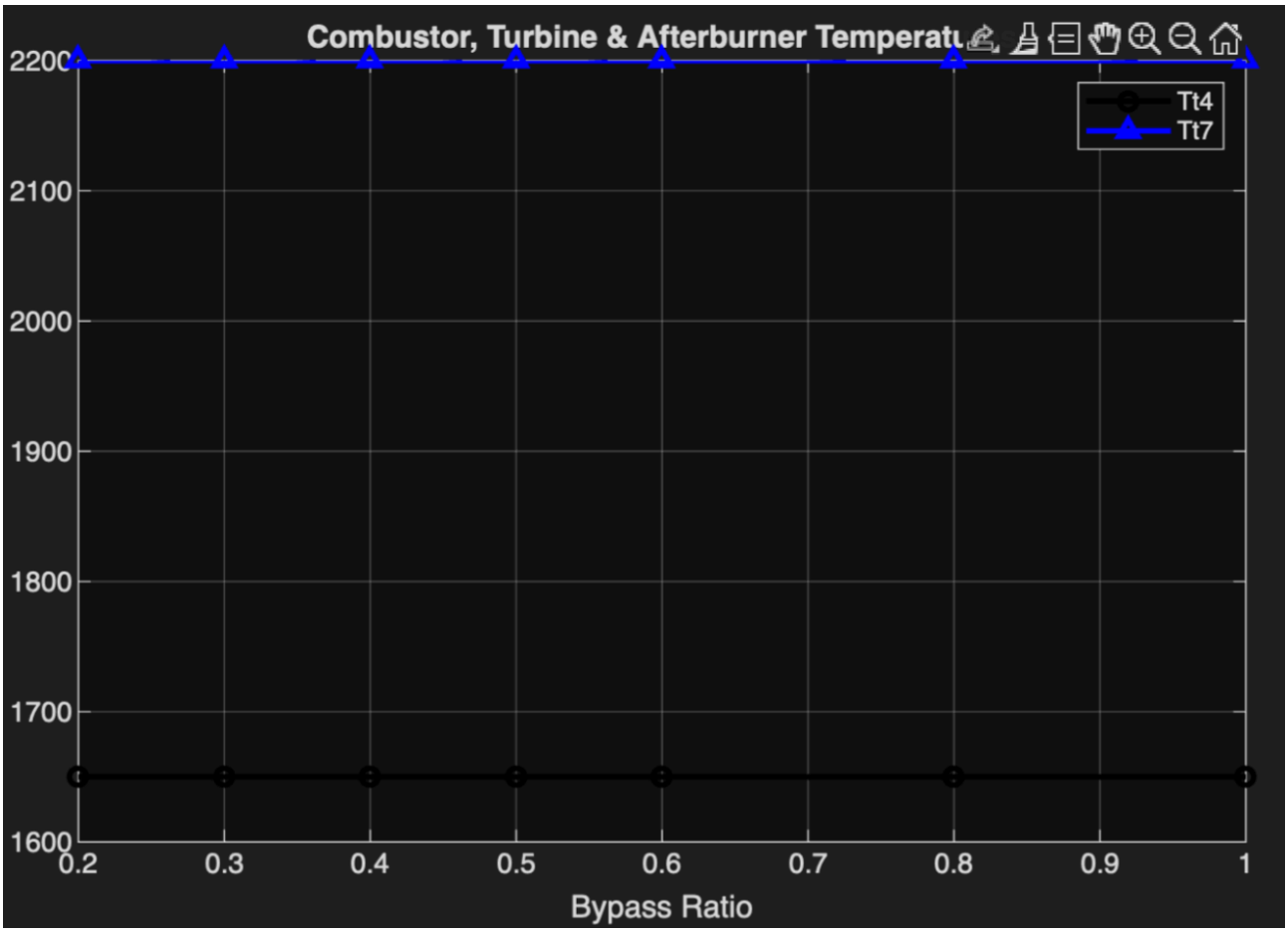

Total Entropy Generation vs Bypass Ratio
1458
1456
1454
1452
1450
1448
1446
1444
1442
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio

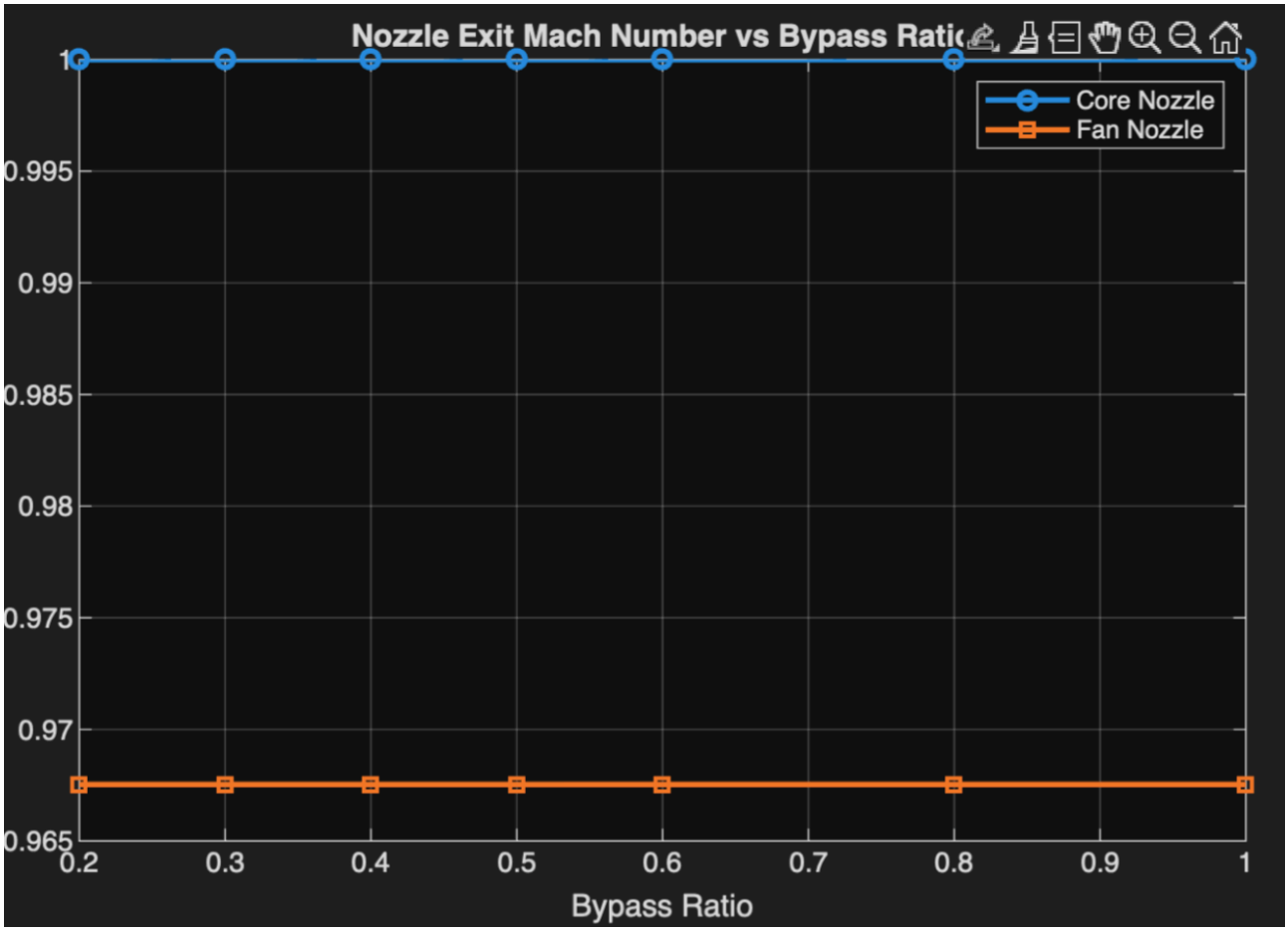

Entropy Generation Across Engine Compone
1000
900
800
700
600
500
400
300
200
100
0
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio
Diffuser
Compressor
Combustor
Turbine
Nozzle

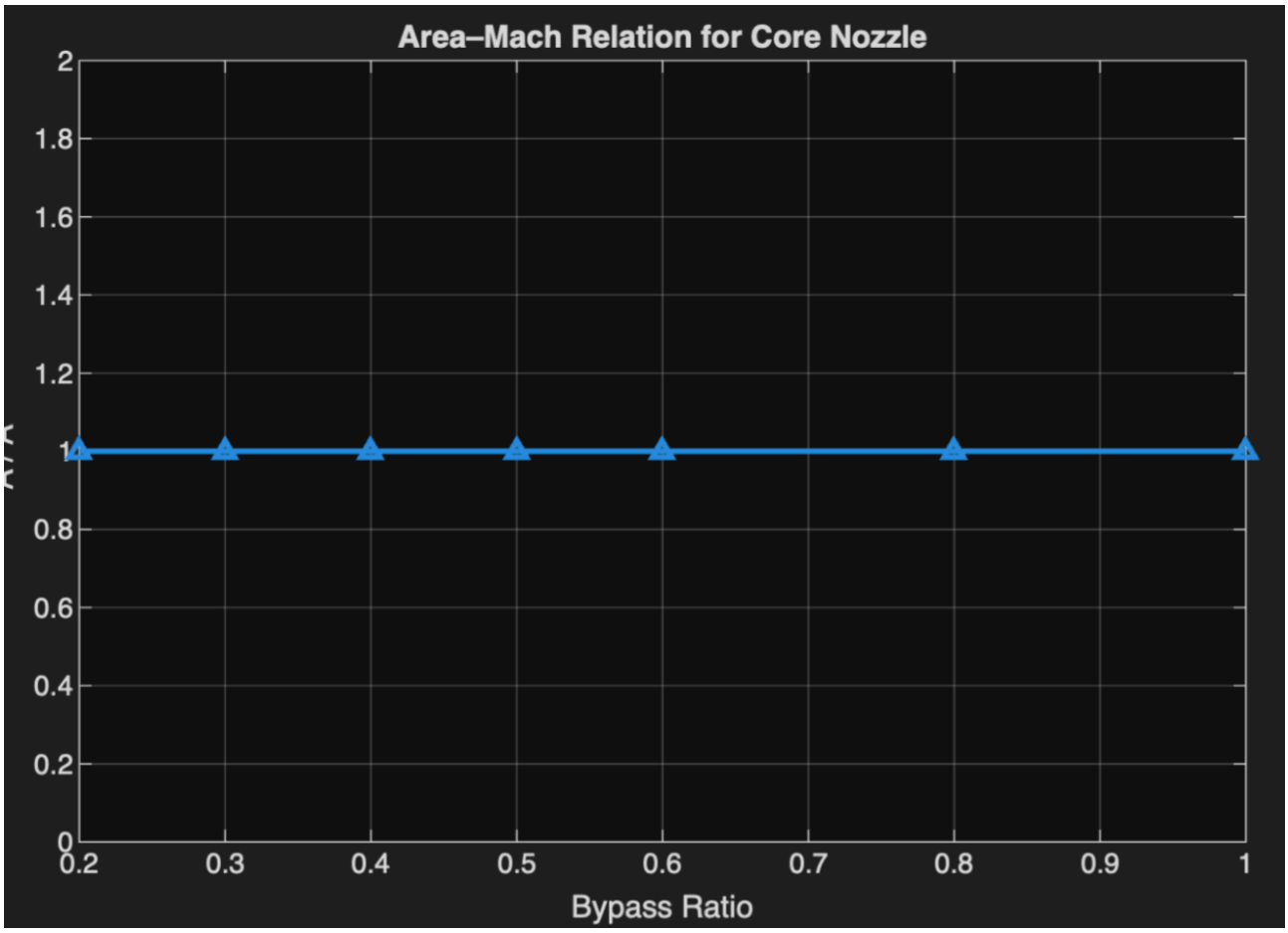

Propulsive Efficiency vs Bypass Ratio
0.62
0.6
0.58
0.56
0.54
0.52
0.5
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio

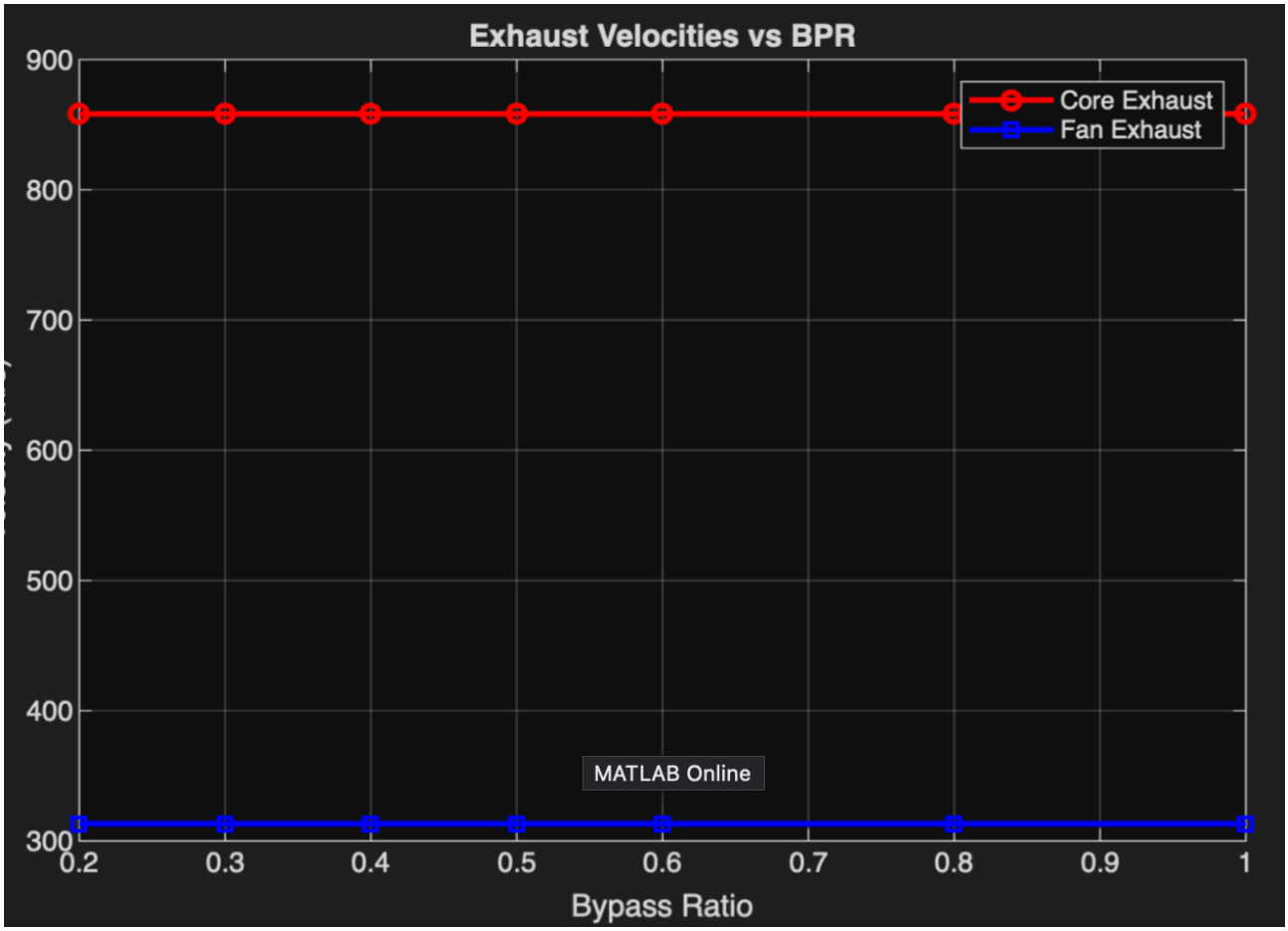

Overall Engine Pressure Ratio vs Bypass Ratio
4.55
4.5
4.45
4.4
4.35
4.3
4.25
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio

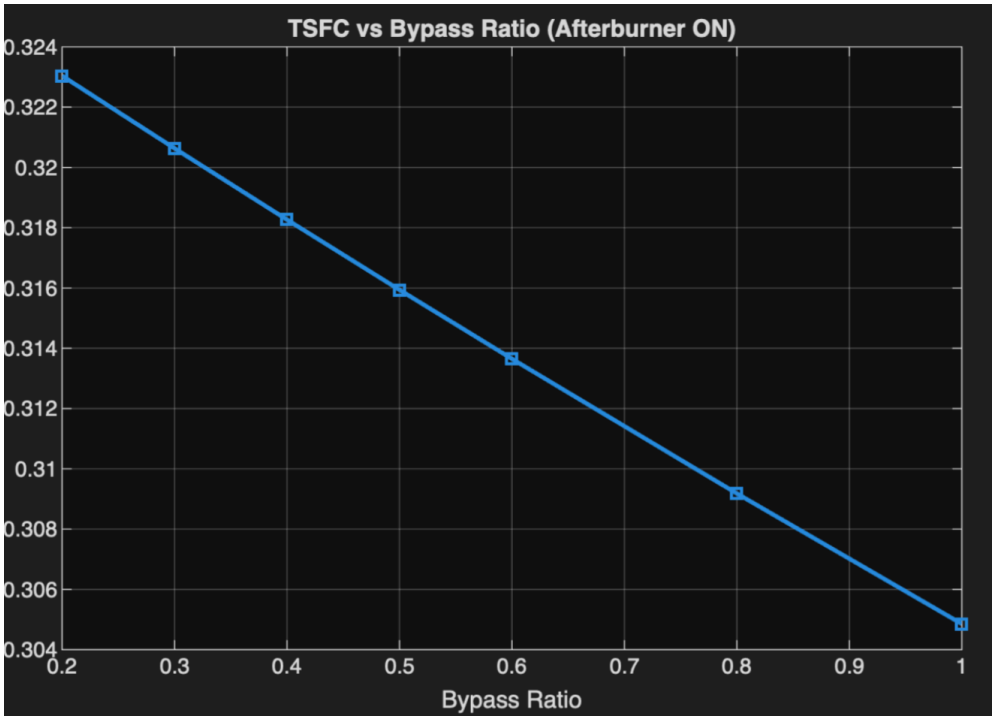

Exhaust Velocity Ratios (V/a0)
Core
Fan
MATLAB Online
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio


Combustor, Turbine & Afterburner Temperatu
Tt4
Tt7
2200
2100
2000
1900
1800
1700
1600
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio


Nozzle Exit Mach Number vs Bypass Ratio
Core Nozzle
Fan Nozzle
1
0.995
0.99
0.985
0.98
0.975
0.97
0.965
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio


Area–Mach Relation for Core Nozzle
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio


Exhaust Velocities vs BPR
Core Exhaust
Fan Exhaust
900
800
700
600
500
400
300
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio
MATLAB Online


TSFC vs Bypass Ratio (Afterburner ON)
0.324
0.322
0.32
0.318
0.316
0.314
0.312
0.31
0.308
0.306
0.304
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio

Froude (Jet) Efficiency vs Bypass Ratio
0.62
0.6
0.58
0.56
0.54
0.52
0.5
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio